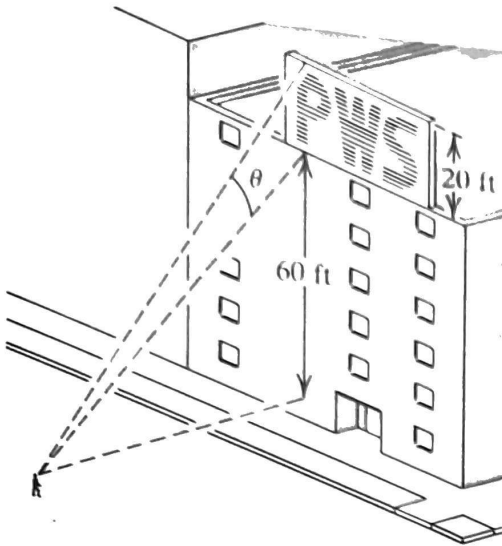
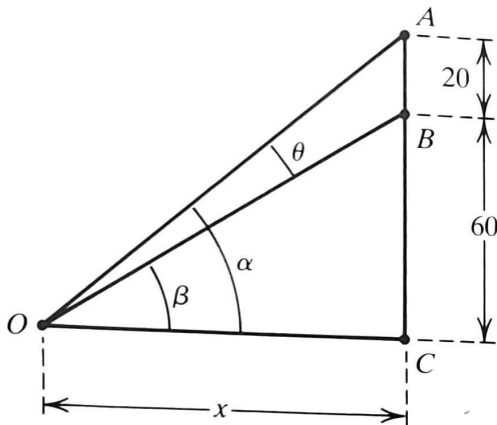


FIGURE 4.52

(i)



(ii)



EXAMPLE 8 A billboard 20 feet tall is located on top of a building, with its lower edge 60 feet above the level of a viewer's eye, as shown in Figure 4.52(i). How far from a point directly below the sign should a viewer stand to maximize the angle θ between the lines of sight of the top and bottom of the billboard? (This angle should result in the best view of the billboard.)

SOLUTION The problem is sketched in Figure 4.52(ii), using right triangles AOC and BOC having common side OC of (variable) length x . We see that

$$\tan \alpha = \frac{80}{x} \quad \text{and} \quad \tan \beta = \frac{60}{x}.$$

The angle $\theta = \alpha - \beta$ is a function of x and

$$\tan \theta = \tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}.$$

Substituting for $\tan \alpha$ and $\tan \beta$ and simplifying, we obtain

$$\tan \theta = \frac{\frac{80}{x} - \frac{60}{x}}{1 + \left(\frac{80}{x}\right)\left(\frac{60}{x}\right)} = \frac{20x}{x^2 + 4800}.$$

The extrema of θ occur if $D_x \theta = 0$. Differentiating implicitly with respect to x and using the quotient rule gives us

$$\sec^2 \theta D_x \theta = \frac{(x^2 + 4800)(20) - 20x(2x)}{(x^2 + 4800)^2} = \frac{96,000 - 20x^2}{(x^2 + 4800)^2}.$$

Since $\sec^2 \theta > 0$, it follows that $D_x \theta = 0$ if and only if

$$96,000 - 20x^2 = 0, \quad \text{or} \quad x^2 = 4800.$$

Thus, the only critical number of θ is

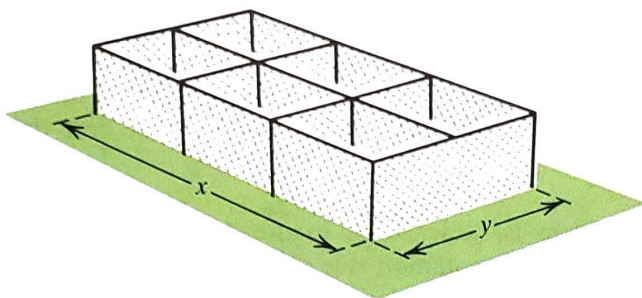
$$x = \sqrt{4800} = 40\sqrt{3}.$$

We may verify that the sign of $D_x \theta$ changes from positive to negative at $\sqrt{4800}$, and hence a maximum value of θ occurs at $x = 40\sqrt{3}$ ft ≈ 69.3 ft.

EXERCISES 4.6

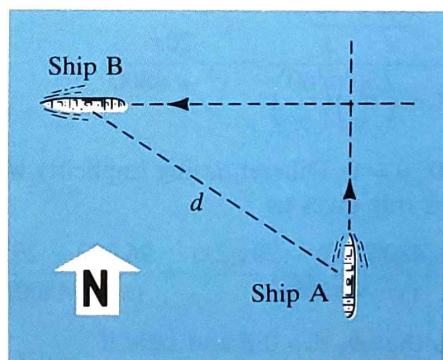
- If a box with a square base and open top is to have a volume of 4 ft^3 , find the dimensions that require the least material. (Disregard the thickness of the material and waste in construction.)
- Work Exercise 1 if the box has a closed top.
- A metal cylindrical container with an open top is to hold one cubic foot. If there is no waste in construction, find the dimensions that require the least amount of material. (Compare with Example 3.)
- If the circular base of the container in Exercise 3 is cut from a square sheet and the remaining metal is discarded, find the dimensions that require the least amount of material.
- One thousand feet of chain link fence will be used to construct six cages for a zoo exhibit, as shown in the figure on the following page. Find the dimensions that maximize the enclosed area A . (Hint: First express y as a function of x , and then express A as a function of x .)

EXERCISE 5



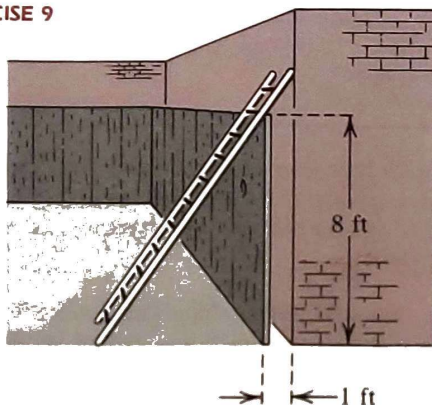
- 6 Refer to Example 6. If the person is in a motorboat that can travel at an average rate of 15 mi/hr, what route should be taken to arrive at the house in the least amount of time?
- 7 At 1:00 P.M. ship A is 30 miles due south of ship B and is sailing north at a rate of 15 mi/hr. If ship B is sailing west at a rate of 10 mi/hr, find the time at which the distance d between the ships is minimal (see figure).

EXERCISE 7



- 8 A window has the shape of a rectangle surmounted by a semicircle. If the perimeter of the window is 15 feet, find the dimensions that will allow the maximum amount of light to enter.
- 9 A fence 8 feet tall stands on level ground and runs parallel to a tall building (see figure). If the fence is 1 foot from

EXERCISE 9



the building, find the length of the shortest ladder that will extend from the ground over the fence to the wall of the building. (Hint: Use similar triangles.)

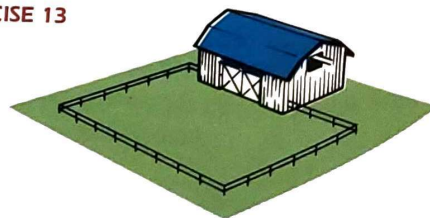
- 10 A page of a book is to have an area of 90 in.^2 , with 1-inch margins at the bottom and sides and a $\frac{1}{2}$ -inch margin at the top. Find the dimensions of the page that will allow the largest printed area.
- 11 A builder intends to construct a storage shed having a volume of 900 ft^3 , a flat roof, and a rectangular base whose width is three-fourths the length. The cost per square foot of the materials is \$4 for the floor, \$6 for the sides, and \$3 for the roof. What dimensions will minimize the cost?
- 12 A water cup in the shape of a right circular cone is to be constructed by removing a circular sector from a circular sheet of paper of radius a and then joining the two straight edges of the remaining paper (see figure). Find the volume of the largest cup that can be constructed.

EXERCISE 12



- 13 A farmer has 500 feet of fencing to enclose a rectangular field. A barn will be used as part of one side of the field (see figure). Prove that the area of the field is greatest when the rectangle is a square.

EXERCISE 13



- 14 Refer to Exercise 13. Suppose the farmer wants the area of the rectangular field to be $A \text{ ft}^2$. Prove that the least amount of fencing is required when the rectangle is a square.
- 15 A hotel that charges \$80 per day for a room gives special rates to organizations that reserve between 30 and 60 rooms. If more than 30 rooms are reserved, the charge per room is decreased by \$1 times the number of rooms over 30. Under these conditions, how many rooms must be rented if the hotel is to receive the maximum income per day?
- 16 Refer to Exercise 15. Suppose that for each room rented it costs the hotel \$6 per day for cleaning and maintenance.

EXERCISE

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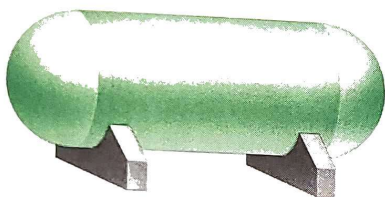
EXERCISES 4.6

217

nance. In this case, how many rooms must be rented to obtain the greatest net income?

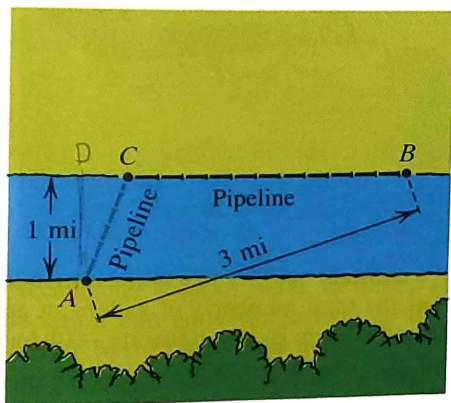
- 17 A steel storage tank for propane gas is to be constructed in the shape of a right circular cylinder with a hemisphere at each end (see figure). The construction cost per square foot for the end pieces is twice that for the cylindrical piece. If the desired capacity is $10\pi \text{ ft}^3$, what dimensions will minimize the cost of construction?

EXERCISE 17



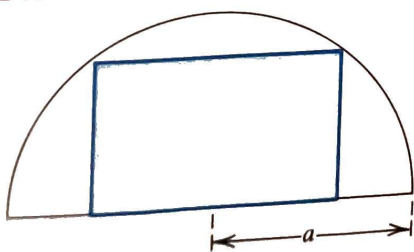
- 18 A pipeline for transporting oil will connect two points A and B that are 3 miles apart and on opposite banks of a straight river 1 mile wide (see figure). Part of the pipeline will run under water from A to a point C on the opposite bank, and then above ground from C to B . If the cost per mile of running the pipeline under water is four times the cost per mile of running it above ground, find the location of C that will minimize the cost (disregard the slope of the river bed).

EXERCISE 18



- 19 Find the dimensions of the rectangle of maximum area that can be inscribed in a semicircle of radius a , if two vertices lie on the diameter (see figure).

EXERCISE 19



- 20 Find the dimensions of the rectangle of maximum area that can be inscribed in an equilateral triangle of side a , if two vertices of the rectangle lie on one of the sides of the triangle.

- 21 Of all possible right circular cones that can be inscribed in a sphere of radius a , find the volume of the one that has maximum volume.

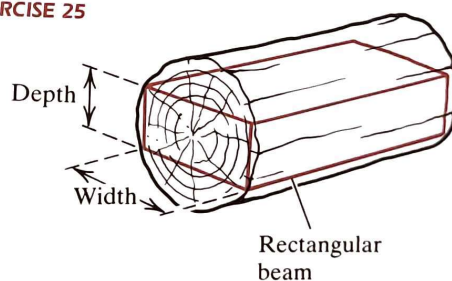
- 22 Find the dimensions of the right circular cylinder of maximum volume that can be inscribed in a sphere of radius a .

- 23 Find the point on the graph of $y = x^2 + 1$ that is closest to the point $(3, 1)$.

- 24 Find the point on the graph of $y = x^3$ that is closest to the point $(4, 0)$.

- 25 The strength of a rectangular beam is directly proportional to the product of its width and the square of the depth of a cross section. Find the dimensions of the strongest beam that can be cut from a cylindrical log of radius a (see figure).

EXERCISE 25



- 26 The illumination from a light source is directly proportional to the strength of the source and inversely proportional to the square of the distance from the source. If two light sources of strengths S_1 and S_2 are d units apart, at what point on the line segment joining the two sources is the illumination minimal?

- 27 A wholesaler sells running shoes at \$20 per pair if less than 50 pairs are ordered. If 50 or more pairs are ordered (up to 600), the price per pair is reduced by 2 cents times the number ordered. What size order will produce the maximum amount of money for the wholesaler?

- 28 A paper cup is to be constructed in the shape of a right circular cone. If the volume desired is $36\pi \text{ in}^3$, find the dimensions that require the least amount of paper. (Disregard any waste that may occur in the construction.)

- 29 A wire 36 centimeters long is to be cut into two pieces. One of the pieces will be bent into the shape of an equilateral triangle and the other into the shape of a rectangle whose length is twice its width. Where should the wire be cut if the combined area of the triangle and rectangle is to be (a) minimized? (b) maximized?

- 30 An isosceles triangle has base b and equal sides of length a . Find the dimensions of the rectangle of maximum area that can be inscribed in the triangle if one side of the rectangle lies on the base of the triangle.
- 31 A window has the shape of a rectangle surmounted by an equilateral triangle. If the perimeter of the window is 12 feet, find the dimensions of the rectangle that will produce the largest area for the window.
- 32 Two vertical poles of lengths 6 feet and 8 feet stand on level ground, with their bases 10 feet apart. Approximate the minimal length of cable that can reach from the top of one pole to some point on the ground between the poles and then to the top of the other pole.
- 33 Prove that the rectangle of largest area having a given perimeter p is a square.
- 34 A right circular cylinder is generated by rotating a rectangle of perimeter p about one of its sides. What dimensions of the rectangle will generate the cylinder of maximum volume?
- 35 The owner of an apple orchard estimates that if 24 trees are planted per acre, then each mature tree will yield 600 apples per year. For each additional tree planted per acre, the number of apples produced by each tree decreases by 12 per year. How many trees should be planted per acre to obtain the most apples per year?
- 36 A real estate company owns 180 efficiency apartments, which are fully occupied when the rent is \$300 per month. The company estimates that for each \$10 increase in rent, 5 apartments will become unoccupied. What rent should be charged in order to obtain the largest gross income?
- 37 A package can be sent by parcel post only if the sum of its length and girth (the perimeter of the base) is not more than 108 inches. Find the dimensions of the box of maximum volume that can be sent, if the base of the box is a square.
- 38 A North-South highway A and an East-West highway B intersect at a point P . At 10:00 A.M. an automobile crosses P traveling north on highway A at a speed of 50 mi/hr. At that same instant, an airplane flying east at a speed of 200 mi/hr and an altitude of 26,400 feet is directly above the point on highway B that is 100 miles west of P . If the automobile and the airplane maintain the same speeds and directions, at what time will they be closest to each other?
- 39 Two factories A and B that are 4 miles apart emit particles in smoke that pollute the area between the factories. Suppose that the number of particles emitted from each factory is directly proportional to the amount of smoke and inversely proportional to the cube of the dis-

tance from the factory. If factory A emits twice as much smoke as factory B, at what point between A and B is the pollution minimal?

- 40 An oil field contains 8 wells, which produce a total of 1600 barrels of oil per day. For each additional well that is drilled, the average production per well decreases by 10 barrels per day. How many additional wells should be drilled to obtain the maximum amount of oil per day?
- 41 A canvas tent is to be constructed in the shape of a pyramid with a square base. A steel pole, placed in the center of the tent, will form the support (see figure). If S ft² of canvas is available for the four sides and x is the length of the base, show that
- (a) the volume V of the tent is $V = \frac{1}{6}x\sqrt{S^2 - x^4}$
- (b) V has its maximum value when x equals $\sqrt[4]{2S}$, the length of the pole

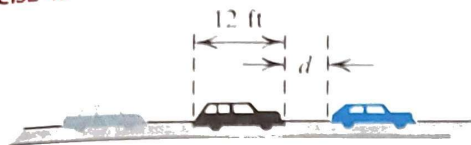
EXERCISE 41



- 42 A boat must travel 100 miles upstream against a 10 mi/hr current. When the boat's velocity is v mi/hr, the number of gallons of gasoline consumed each hour is directly proportional to v^2 .
- (a) If a constant velocity of v mi/hr is maintained, show that the total number y of gallons of gasoline consumed is given by $y = 100kv^2/(v - 10)$ for $v > 10$ and for some positive constant k .
- (b) Find the speed that minimizes the number of gallons of gasoline consumed during the trip.
- 43 Cars are crossing a bridge that is 1 mile long. Each car is 12 feet long and is required to stay a distance of at least d feet from the car in front of it (see figure).
- (a) Show that the largest number of cars that can be on the bridge at one time is $\llbracket 5280/(12 + d) \rrbracket$, where $\llbracket \cdot \rrbracket$ denotes the greatest integer function.
- (b) If the velocity of each car is v mi/hr, show that the maximum traffic flow rate F (in cars/hr) is given by $F = \llbracket 5280v/(12 + d) \rrbracket$.
- (c) The stopping distance (in feet) of a car traveling v mi/hr is approximately $0.05v^2$. If $d = 0.025v^2$, find

the speed that maximizes the traffic flow across the bridge.

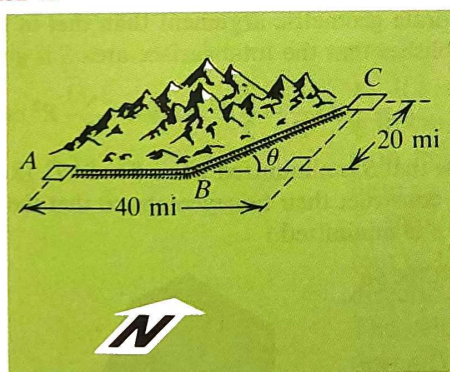
EXERCISE 43



- 44 Prove that the shortest distance from a point (x_1, y_1) to the graph of a differentiable function f is measured along a normal line to the graph—that is, a line perpendicular to the tangent line.

- 45 A railroad route is to be constructed from town A to town C, branching out from a point B toward C at an angle of θ degrees (see figure). Because of the mountains between A and C, the branching point B must be at least 20 miles east of A. If the construction costs are \$50,000 per mile between A and B and \$100,000 per mile between B and C, find the branching angle θ that minimizes the total construction cost.

EXERCISE 45



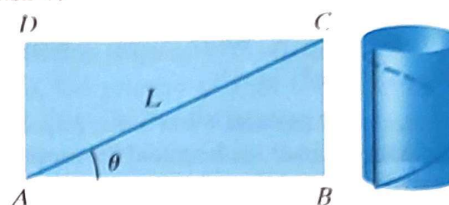
- 46 A long rectangular sheet of metal, 12 inches wide, is to be made into a rain gutter by turning up two sides at angles of 120° to the sheet. How many inches should be turned up to give the gutter its greatest capacity?

- 47 Refer to Exercise 12. Find the central angle of the sector that will maximize the volume of the cup.

- 48 A square picture having sides 2 feet long is hung on a wall such that the base is 6 feet above the floor. If a person whose eye level is 5 feet above the floor looks at the picture and if θ is the angle between the line of sight and the top and bottom of the picture, find the person's distance from the wall at which θ has its maximum value.

- 49 A rectangle made of elastic material will be made into a cylinder by joining edges AD and BC (see figure). To support the structure, a wire of fixed length L is placed along the diagonal of the rectangle. Find the angle θ that will result in the cylinder of maximum volume.

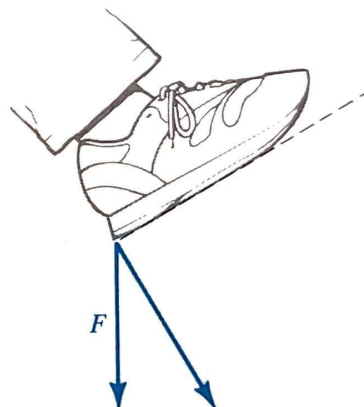
EXERCISE 49



- 50 When a person is walking, the magnitude F of the vertical force of one foot on the ground (see figure) can be approximated by $F = A(\cos bt - a \cos 3bt)$ for time t (in seconds), with $A > 0$, $b > 0$, and $0 < a < 1$.

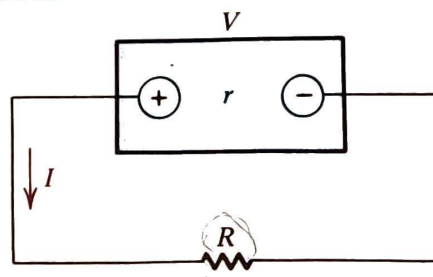
- (a) Show that $F = 0$ when $t = -\pi/(2b)$ and $t = \pi/(2b)$. (The time $t = -\pi/(2b)$ corresponds to the instant when the foot first touches the ground and the weight of the body is being supported by the other foot.)
- (b) Show that the maximum force occurs when $t = 0$ or when $\sin^2 bt = (9a - 1)/(12a)$.
- (c) If $a = \frac{1}{3}$, express the maximum force in terms of A .
- (d) If $0 < a \leq \frac{1}{9}$, express the maximum force in terms of A .

EXERCISE 50



- 51 A battery having fixed voltage V and fixed internal resistance r is connected to a circuit that has variable resistance R (see figure). By Ohm's law, the current I in the circuit is $I = V/(R + r)$. If the power output P is given by $P = I^2 R$, show that the maximum power occurs if $R = r$.

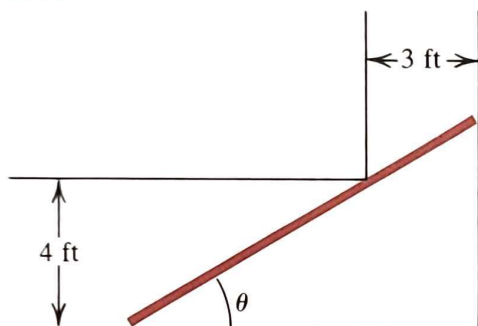
EXERCISE 51



- 52 The power output P of an automobile battery is given by $P = VI - I^2r$ for voltage V , current I , and internal resistance r of the battery. What current corresponds to the maximum power?

- 53 Two corridors 3 feet and 4 feet wide, respectively, meet at a right angle. Find the length of the longest nonbendable rod that can be carried horizontally around the corner, as shown in the figure. (Disregard the thickness of the rod.)

EXERCISE 53

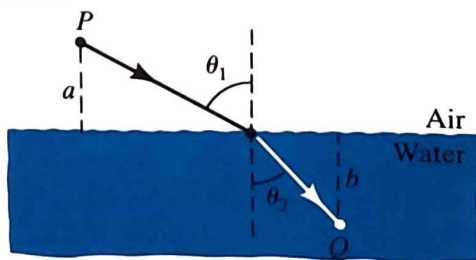


- 54 Light travels from one point to another along the path that requires the least amount of time. Suppose that light has velocity v_1 in air and v_2 in water, where $v_1 > v_2$. If light travels from a point P in air to a point Q in water (see figure), show that the path requires the least amount of time if

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{v_1}{v_2}.$$

(This is an example of Snell's law of refraction.)

EXERCISE 54



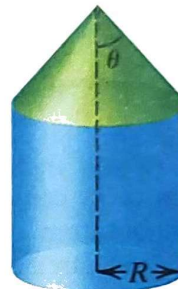
- 55 A circular cylinder of fixed radius R is surmounted by a cone (see figure). The ends of the cylinder are open, and the total volume is to be a specified constant V .

(a) Show that the total surface area S is given by

$$S = \frac{2V}{R} + \pi R^2 \left(\csc \theta - \frac{2}{3} \cot \theta \right).$$

(b) Show that S is minimized when $\theta \approx 48.2^\circ$.

EXERCISE 55



- 56 In the classic honeycomb-structure problem, a hexagonal prism of fixed radius (and side) R is surmounted by adding three identical rhombuses that meet in a common vertex (see figure). The bottom of the prism is open, and the total volume is to be a specified constant V . A more elaborate geometric argument than that in Exercise 55 establishes that the total surface area S is given by

$$S = \frac{4}{3} \sqrt{3} \frac{V}{R} - \frac{3}{2} R^2 \cot \theta + \frac{3\sqrt{3}}{2} R^2 \csc \theta.$$

Show that S is minimized when $\theta \approx 54.7^\circ$. (Remarkably, bees construct their honeycombs so that the amount of wax S is minimized.)

EXERCISE 56



4.7 RECTILINEAR MOTION AND OTHER APPLICATIONS

In this section we use derivatives to describe several important types of motion that occur in physical situations. We also discuss how derivatives may be applied to problems in economics.